

Umair Mehfooz

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SUMMARY

Artificial Intelligence undergraduate and Machine Learning Intern at FlyRank AI who builds and ships production software end to end: two live e-commerce stores and an offline business-management app for clients, plus LLM applications with LangChain and FastAPI. Works across Python, TypeScript, Next.js, Supabase/PostgreSQL and data tooling such as Pandas and DuckDB.

EDUCATION

Capital University of Science and Technology (CUST) | Islamabad, Pakistan **2024 – 2028 (Expected)**
Bachelor of Science in Artificial Intelligence (BSAI) – Dean's Honor Roll (4 times)

Federal Government Sir Syed College | Rawalpindi, Pakistan **2022 – 2023**
Intermediate (FSc Pre-Engineering)

EXPERIENCE

FlyRank AI – Machine Learning Intern **Jun 2026 – Present**

- Analyzing the signals behind Google search ranking and content discoverability using FlyRank's anonymized search data.
- Querying a ~79M-row search dataset with DuckDB directly from Google Colab, without local downloads.
- Running an end-to-end ML workflow: task framing, data contracts, leakage checks, signal audits, baselines and validation.

Freelance Full-Stack Developer – Independent Client Work **Aug 2026 – Present**

- Naavya** (naavya.online): Launched a production fashion e-commerce store on Next.js, TypeScript and Supabase (PostgreSQL, Auth, RLS) with COD and Safepay checkout, customer accounts, order emails and an admin dashboard.
- Ideal Jewellers** (idealjewellers.pk): Built a luxury jewellery storefront with category filters, bridal collections, search, wishlist and checkout; ported the static prototype to Next.js with the catalogue and admin area on Supabase.
- Aqua Aman**: Built an offline Windows desktop app (Electron, React, SQLite) that runs a water-delivery business: deliveries, payments, bottle stock, P&L and receivables reports, scheduled WhatsApp receipts, automatic daily backups and a PIN lock.

PROJECTS

KeepMe – AI Accountability Coach | Next.js, FastAPI, Supabase, pgvector, OpenRouter, Docker **2026**

- Built a full-stack AI coaching PWA that tracks commitments, extracts structured facts from daily reflections, scores progress, and sends reports and Web Push reminders.
- Computed coaching tone deterministically so the LLM picks the wording, never the severity; replies pass a safety filter.
- Enforced PostgreSQL Row Level Security through FastAPI by verifying Supabase JWTs per request; added PII redaction, AI usage quotas, a circuit breaker and model fallback; deployed with Docker and Caddy.

BidForge AI – RAG Proposal Engine (Hackathon) | LangChain, FAISS, FastAPI, Groq API **Jun 2026**

- Built a RAG pipeline matching RFP requirements to capabilities using MiniLM embeddings, FAISS and re-ranking.
- Generated compliance verdicts and proposal drafts with Groq LLMs, cutting drafting time from hours to minutes.

Portal Automation Framework | Python, Playwright, Pytest **Jun 2026**

- Built a browser automation framework with the Page Object Model, session reuse, parallel tests and failure screenshots.

TECHNICAL SKILLS

Programming Languages: Python, TypeScript, JavaScript, SQL, C++

AI/ML & Data Science: NumPy, Pandas, Matplotlib, Scikit-learn, DuckDB, Jupyter, Machine Learning, RAG

AI & LLM Frameworks: LangChain, Prompt Engineering, LLM Application Development, OpenRouter, Groq API, pgvector

Web & Backend: Next.js, React, Tailwind CSS, FastAPI, REST APIs, Electron

Databases: PostgreSQL, Supabase, MongoDB, SQLite

Tools & Cloud: Git, GitHub, Docker, Vercel, Playwright, Oracle Cloud Infrastructure (OCI), MS Excel

CERTIFICATIONS

- Oracle Cloud Infrastructure (OCI) Certified Foundations Associate – Oracle **Aug 2025**
- Python for Data Analysis: Pandas & NumPy – Coursera

LANGUAGES & INTERESTS

Languages: English, Urdu, German (A1 certificate, NUML)

Interests: Book Reading, Gardening, Socializing